

Module 7: MariaDB Assignment

Problem Statement:

You work for XYZ Corporation. Their application requires a SQL service that can store data which can be retrieved if required. Implement a suitable RDS engine for the same. While migrating, you are asked to perform the following tasks:

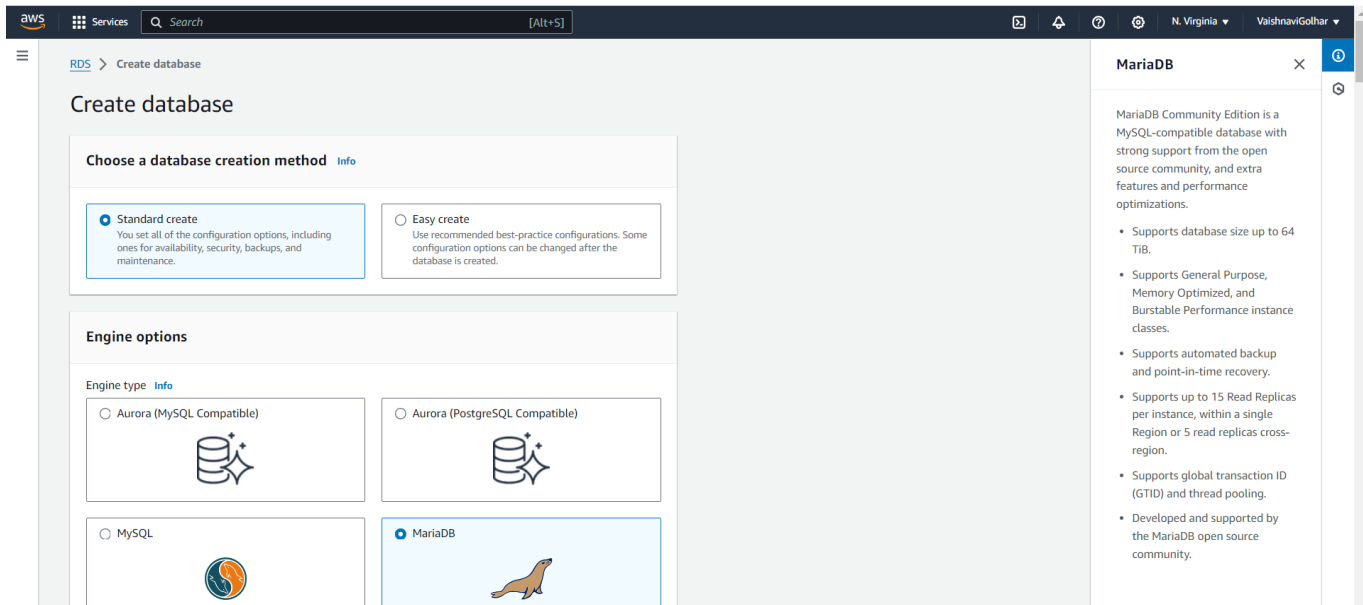
1. Create a MariaDB Engine based RDS Database.
2. Connect to the DB using the following ways:
 - a. SQL Client for Windows
 - b. Linux based EC2 instance

SOLUTION:

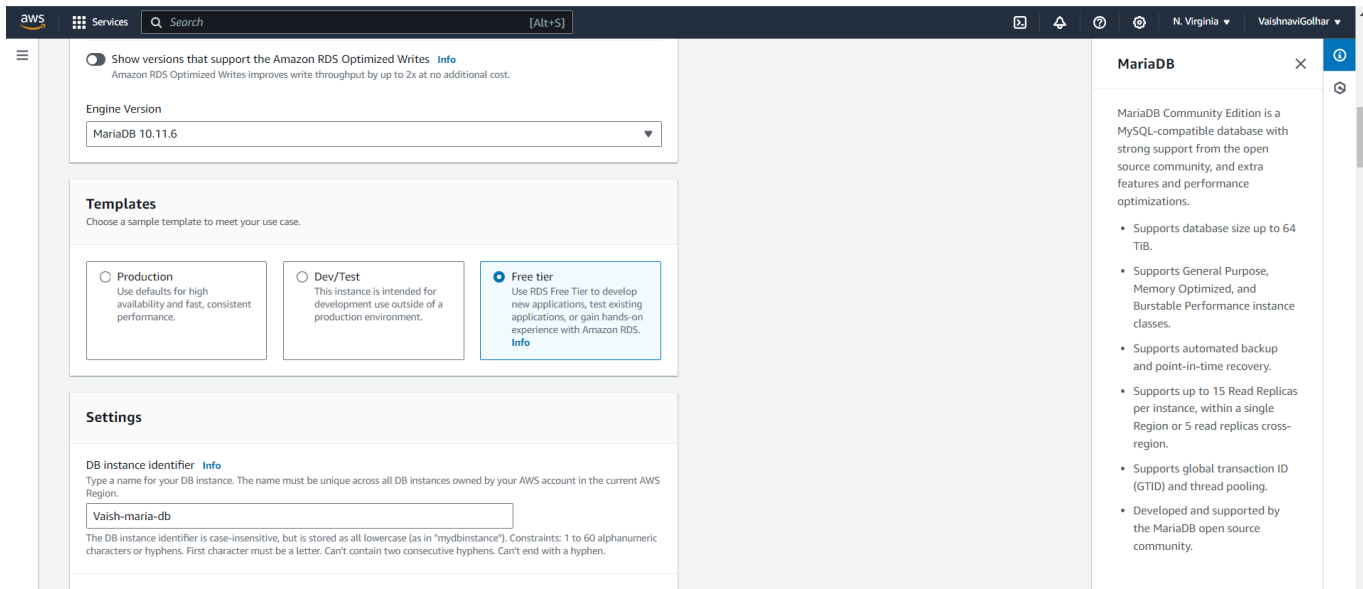
Step 1: Create a MariaDB Engine Based RDS Database

1. Log in to the AWS Management Console and open the Amazon RDS console at <https://console.aws.amazon.com/rds/>.
2. Create Database: Click on "Create database". In the "Choose a database creation method" section, make sure "Standard Create" is selected.

Engine Options: Select "MariaDB" as the engine type.



3. Templates: Choose a template that suits your use case (Free tier, Dev/Test, or Production).



4. Settings: Provide a DB instance identifier, Master username, and Master password. Remember these credentials as you'll need them to connect to the database later.

Region: Vaish-maria-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

☐ **Manage master credentials in AWS Secrets Manager**
Manage master user credentials in Secrets Manager. RDS can generate a password for you and manage it throughout its lifecycle.

[ⓘ](#) If you manage the master user credentials in Secrets Manager, some RDS features aren't supported. [Learn more](#)

☐ **Auto generate a password**
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Constraints: At least 8 printable ASCII characters. Can't contain any of the following: / (slash), ' (single quote), " (double quote) and @ (at sign).

Confirm master password [Info](#)

MariaDB

MariaDB Community Edition is a MySQL-compatible database with strong support from the open source community, and extra features and performance optimizations.

- Supports database size up to 64 TiB.
- Supports General Purpose, Memory Optimized, and Burstable Performance instance classes.
- Supports automated backup and point-in-time recovery.
- Supports up to 15 Read Replicas per instance, within a single Region or 5 read replicas cross-region.
- Supports global transaction ID (GTID) and thread pooling.
- Developed and supported by the MariaDB open source community.

5. DB Instance Size: Select the appropriate instance size.

6. Storage: Configure your storage settings as needed.

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☒ **Show instance classes that support Amazon RDS Optimized Writes** [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ **Include previous generation classes**

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ **Burstable classes (includes t classes)**

db.t3.micro
2 vCPUs 1 GiB RAM Network: 2,085 Mbps

Storage

Storage type [Info](#)
General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage [Info](#)
20 GiB

The minimum value is 20 GiB and the maximum value is 6,144 GiB

7. Connectivity: Under Network & Security, make sure to select a VPC, enable public access if you need to connect from outside AWS (e.g., your local machine), and configure the VPC security group(s) to allow inbound traffic on the default MariaDB port (3306).

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Services

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Availability & durability

Multi-AZ deployment

Info

☒ Create a standby instance (recommended for production usage)

Creates a standby in a different Availability Zone (AZ) to provide data redundancy, eliminate I/O freezes, and minimize latency spikes during system backups.

☐ Do not create a standby instance

Connectivity

Info

Don't connect to an EC2 compute resource

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource

Set up a connection to an EC2 compute resource for this database.

Network type

Info

☒ IPv4

Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode

Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC)

Info

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-0b22f56ff603e7fa1)

6 Subnets, 6 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

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Creating an RDS security group

Choose one or more options

default

Availability Zone [Info](#)

No preference

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ Create an RDS Proxy [Info](#)

RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)

Expiry: May 26, 2061

If you don't select a certificate authority, RDS chooses one for you.

Additional configuration

Database port [Info](#)

TCP/IP port that the database will use for application connections.

3306

Database authentication

8. Database Authentication: Choose “Password authentication”.

Database authentication

Database authentication options [Info](#)

☒ Password authentication
Authenticates using database passwords.

☐ Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.

Monitoring

☐ Enable Enhanced Monitoring
Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

▼ Additional configuration
Database options, encryption turned off, backup turned off, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name [Info](#)

If you do not specify a database name, Amazon RDS does not create a database.

9. Additional Configuration: Provide an initial database name under “Initial database name” if desired. You can leave the other options at their default settings or adjust them according to your requirements.

▼ Additional configuration
Database options, encryption turned off, backup turned off, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name [Info](#)

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group [Info](#)

Option group [Info](#)

Backup

☐ Enable automated backups
Creates a point-in-time snapshot of your database

Encryption

☐ Enable encryption
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. [Info](#)

10. Create Database: Review all settings and click “Create database”. Wait for the database to become available.

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DB instance

12.41 USD

Storage

2.30 USD

Total

14.71 USD

This billing estimate is based on on-demand usage as described in [Amazon RDS Pricing](#). Estimate does not include costs for backup storage, IOs (if applicable), or data transfer.

Estimate your monthly costs for the DB Instance using the [AWS Simple Monthly Calculator](#).

Estimated monthly costs

The Amazon RDS Free Tier is available to you for 12 months. Each calendar month, the free tier will allow you to use the Amazon RDS resources listed below for free:

- 750 hrs of Amazon RDS in a Single-AZ db.t2.micro, db.t3.micro or db.t4g.micro Instance.
- 20 GB of General Purpose Storage (SSD).
- 20 GB for automated backup storage and any user-initiated DB Snapshots.

[Learn more about AWS Free Tier](#).

When your free usage expires or if your application use exceeds the free usage tiers, you simply pay standard, pay-as-you-go service rates as described in the [Amazon RDS Pricing page](#).

You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

Cancel

Create database

▼ Storage autoscaling

Storage autoscaling

Info

Provides dynamic scaling support for your database's storage based on your application's needs.

☐ **Enable storage autoscaling**
Enabling this feature will allow the storage to increase after the specified threshold is exceeded.

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Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Creating database vaish-maria-db

View credential details

Your database might take a few minutes to launch.

You can use settings from vaish-maria-db to simplify configuration of suggested database add-ons while we finish creating your DB for you.

RDS > Databases

Consider creating a Blue/Green Deployment to minimize downtime during upgrades

You may want to consider using Amazon RDS Blue/Green Deployments and minimize your downtime during upgrades. A Blue/Green Deployment provides a staging environment for changes to production databases. [RDS User Guide](#) [Aurora User Guide](#)

Databases (1)

Group resources

Modify

Actions

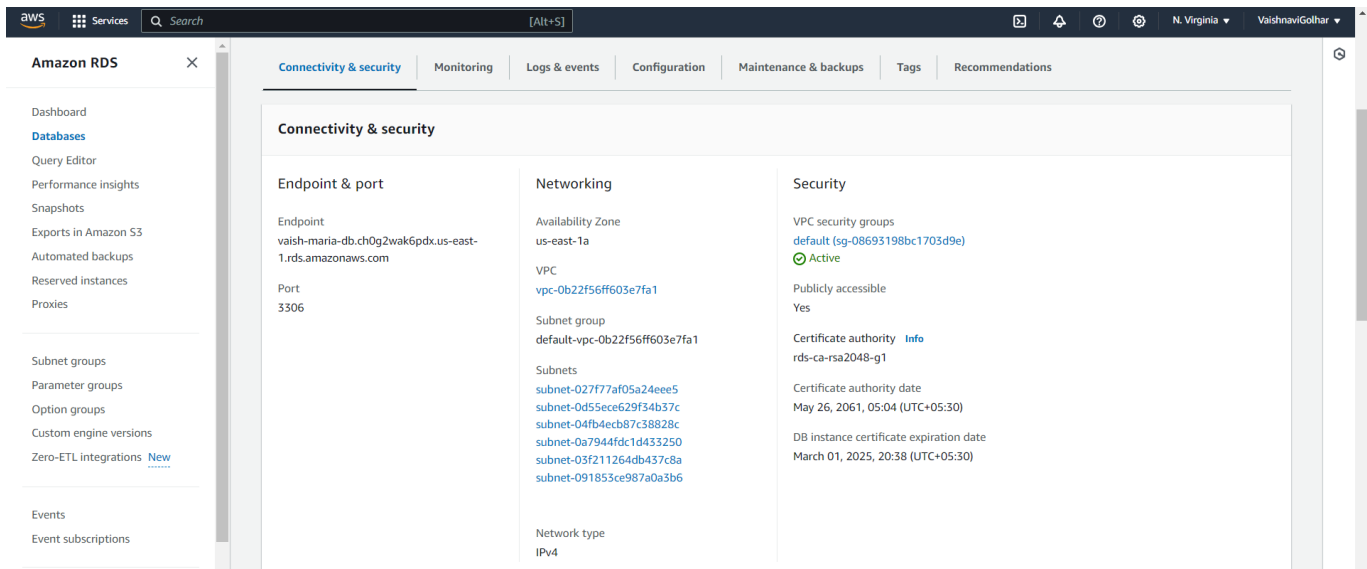
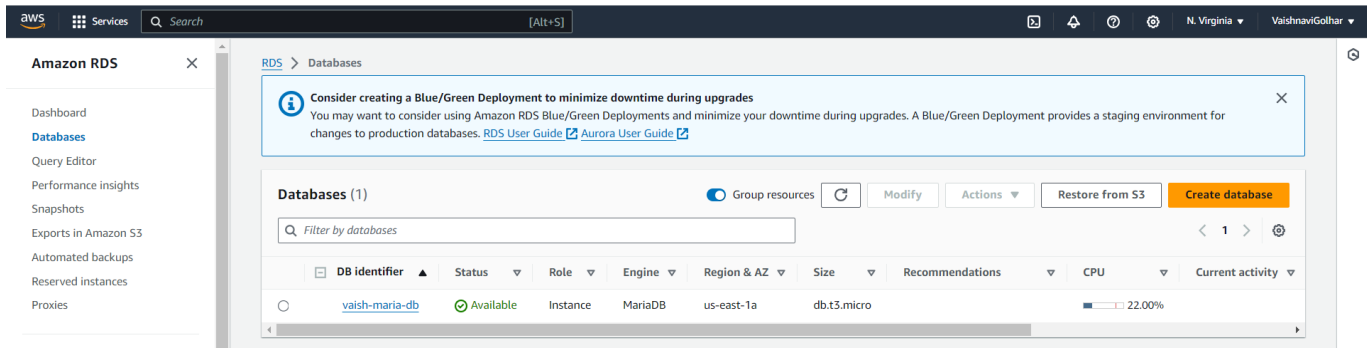
Restore from S3

Create database

Filter by databases

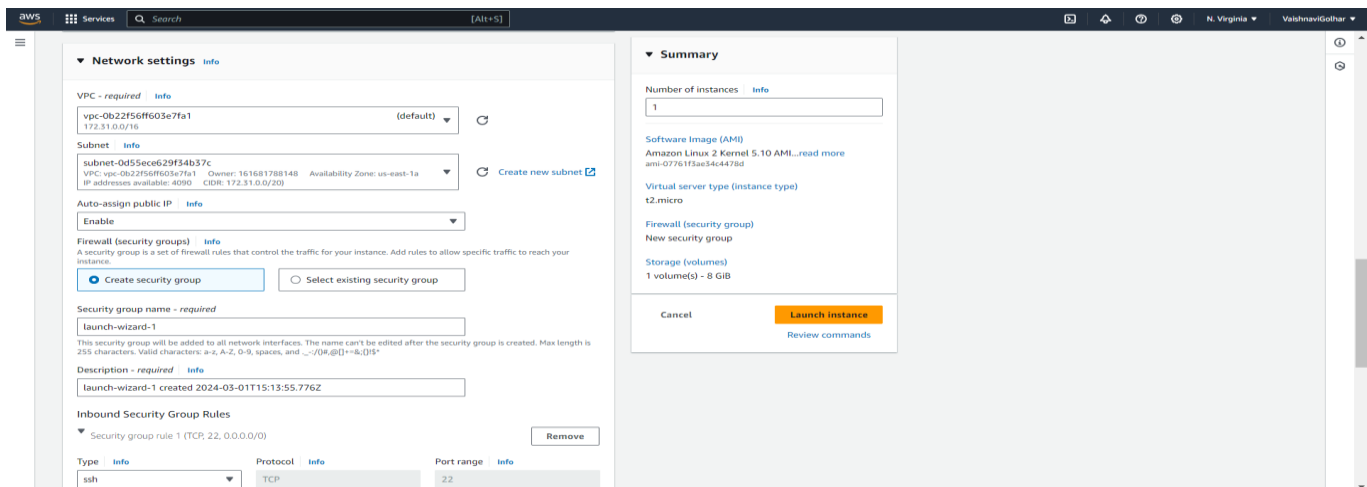
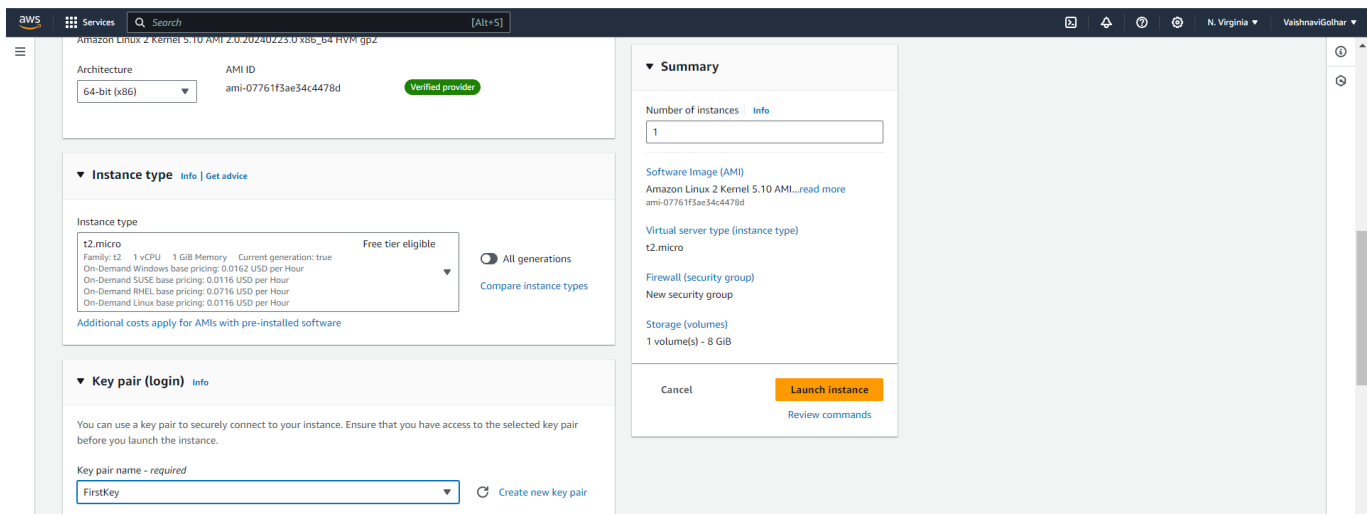
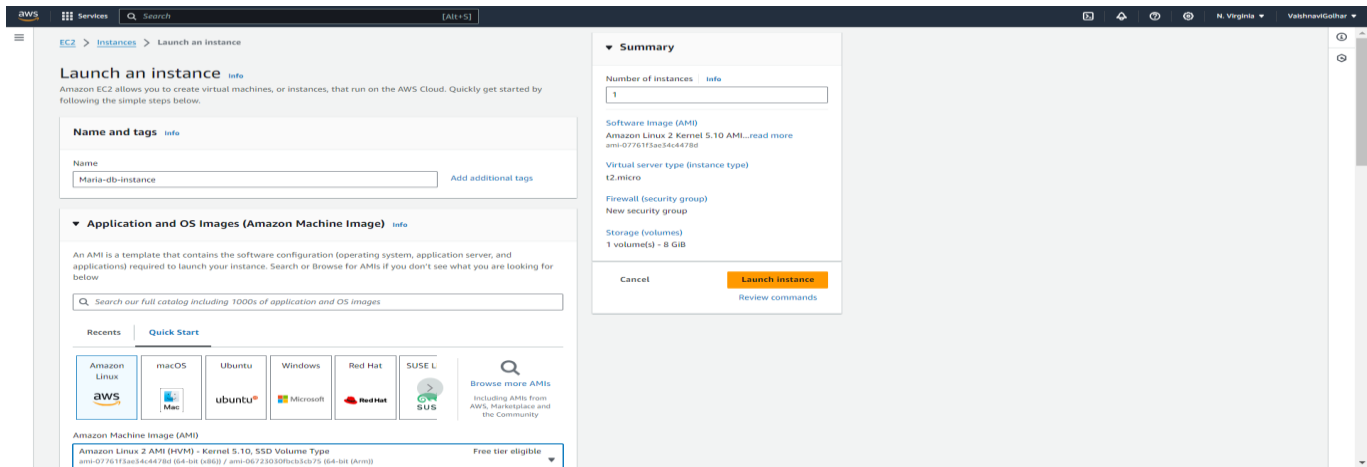
< 1 >

DB identifier	Status	Role	Engine	Region & AZ	Size	Recommendations	CPU	Current activity	Maint
vaish-maria-db	Creating	Instance	MariaDB	-	db.t3.micro	-	-	-	none



b. Linux based EC2 Instance

1. To connect from a Linux EC2 instance, you need to ensure the instance is in the same VPC as the RDS instance or has appropriate routes and security groups set up to allow access.



launch-wizard-1

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-:/[!@#%&'()*+,-.{}~]*

Description - *required* [Info](#)

launch-wizard-1 created 2024-03-01T15:13:55.776Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type [Info](#) Protocol [Info](#) Port range [Info](#)

ssh TCP 22

Source type [Info](#) Source [Info](#) Description - optional [Info](#)

Anywhere 0.0.0.0/0 e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

► Advanced network configuration

Configure storage [Info](#) [Advanced](#)

1x 8 GIB gp2 Root volume (Not encrypted)

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2 Kernel 5.10 AMI...[read more](#)

ami-07761f3ae34c4478d

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GIB

Cancel Launch instance

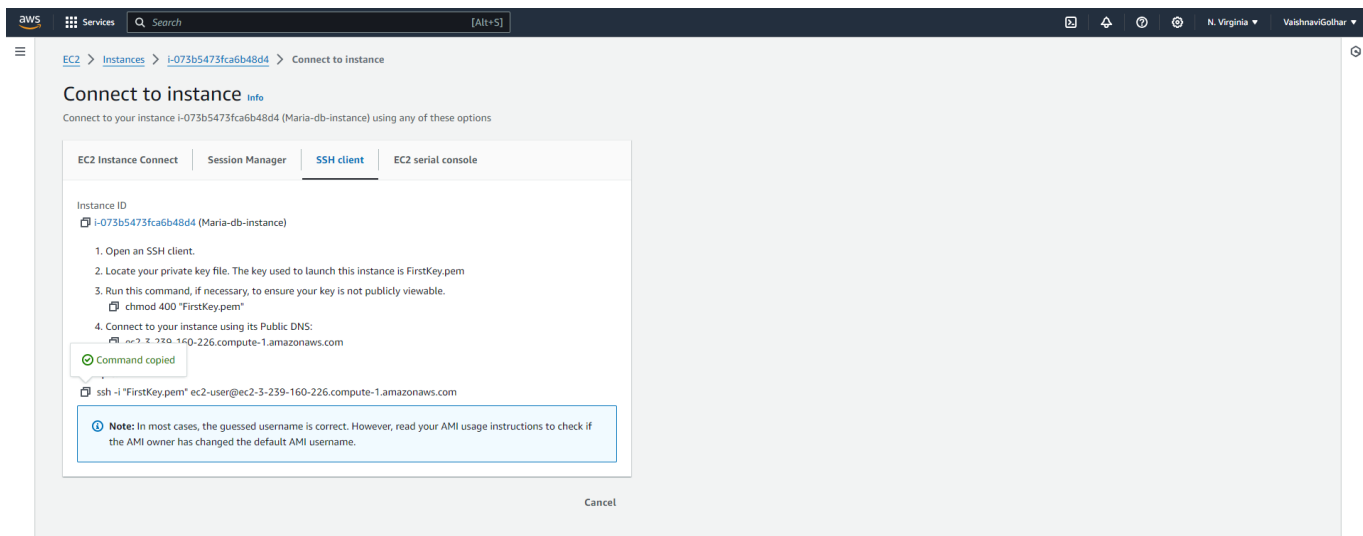
[Review commands](#)

Instances (1) [Info](#)

Find Instance by attribute or tag (case-sensitive) Any state

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IF
☐	Maria-db-insta...	i-073b5473fca6b48d4	Running	t2.micro	Initializing	View alarms	us-east-1a	ec2-3-239-160-226.co...	3.239.160.226	-	-

2. SSH into Your EC2 Instance: Use your terminal or SSH client to connect to your Linux EC2 instance.



Install MariaDB Client: You may need to install the MariaDB client. On Amazon Linux 2, you can do this by running:

```
ec2-user@ip-172-31-14-29:~
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Suttu>cd downloads

C:\Users\Suttu\Downloads>sudo ssh -i "FirstKey.pem" ec2-user@ec2-3-239-160-226.compute-1.amazonaws.com
'sudo' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Suttu\Downloads>sudo su
'sudo' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Suttu\Downloads>ssh -i "FirstKey.pem" ec2-user@ec2-3-239-160-226.compute-1.amazonaws.com
The authenticity of host 'ec2-3-239-160-226.compute-1.amazonaws.com (3.239.160.226)' can't be established.
ECDSA key fingerprint is SHA256:mb4CyyRumQdIFd901gP602bJfi6itiTQ34+P9wMki0.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-239-160-226.compute-1.amazonaws.com,3.239.160.226' (ECDSA) to the list of known hosts.

#
#####
AL2 End of Life is 2025-06-30.

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

#
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https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-172-31-14-29 ~]$
```

```
[ec2-user@ip-172-31-14-29 ~]$ sudo yum update
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
No packages marked for update
[ec2-user@ip-172-31-14-29 ~]$ sudo yum install mariadb
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
Resolving Dependencies
--> Running transaction check
--> Package mariadb.x86_64 1:5.5.68-1.amzn2.0.1 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package                               Arch                               Version                               Repository                               Size
=====
Installing:
mariadb                               x86_64                             1:5.5.68-1.amzn2.0.1                 amzn2-core                               8.8 M

Transaction Summary
=====
Install 1 Package

Total download size: 8.8 M
Installed size: 49 M
Is this ok [y/d/N]: y
Downloading packages:
mariadb-5.5.68-1.amzn2.0.1.x86_64.rpm | 8.8 MB 00:00:00
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Installing : 1:mariadb-5.5.68-1.amzn2.0.1.x86_64 1/1
  Verifying  : 1:mariadb-5.5.68-1.amzn2.0.1.x86_64 1/1

Installed:
  mariadb.x86_64 1:5.5.68-1.amzn2.0.1

Complete!
[ec2-user@ip-172-31-14-29 ~]$
```

```

AWS Services Q Search [Alt+S] N. Virginia VaishnaviGolhar
Last login: Fri Mar 1 15:21:39 2024 from 223.233.84.30
#
_ _ _ _ _
/ _ _ _ _ \   Amazon Linux 2
/_ _ _ _ _ \
/_ _ _ _ _ \   AL2 End of Life is 2025-06-30.
/_ _ _ _ _ \
/_ _ _ _ _ \   A newer version of Amazon Linux is available!
/_ _ _ _ _ \
/_ _ _ _ _ \   Amazon Linux 2023, GA and supported until 2028-03-15.
/_ _ _ _ _ \   https://aws.amazon.com/linux/amazon-linux-2023/
/_ _ _ _ _ \

[ec2-user@ip-172-31-14-29 ~]$ mysql -h vaish-maria-db.ch0g2wak6pdx.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 47
Server version: 10.11.6-MariaDB managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

a. SQL Client for Windows

1. Download and Install HeidiSQL: If you haven't already installed HeidiSQL, you can download it from the official website: [HeidiSQL Download Page](#).
2. Launch HeidiSQL: After installation, launch HeidiSQL from your Windows machine.
3. Create a New Session:
4. Click on "Session" in the menu bar.

5. Select "New Session" from the dropdown menu.

6. Configure Connection Settings:

In the "Settings" tab:

Network Type: Select "MariaDB or MySQL (TCP/IP)".

Hostname/IP: Enter the endpoint of your RDS instance. You can find this on the RDS dashboard under the "Connectivity & Security" tab of your database.

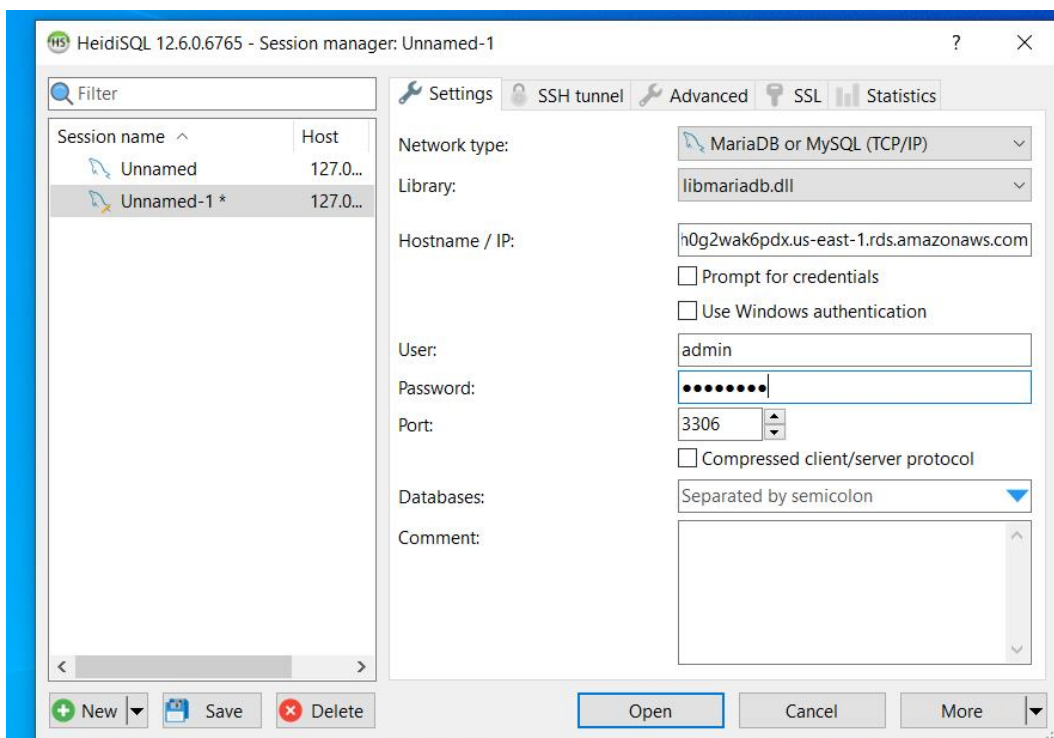
User: Enter the master username you set during the database creation.

Password: Enter the password associated with the master username.

Port: The default port for MariaDB is 3306.

You can leave other settings as default unless you have specific requirements.

Test Connection: After entering the connection details, you can click on the "Open" button to test the connection. If everything is set up correctly, HeidiSQL should establish a connection to your RDS instance.



7. Save the Session: If the test is successful, you can save the session for future use by clicking on "File" > "Save Session As". Give your session a name and click "Save".

8. Connect to the Database: Once the session is saved, you can simply double-click on it in the session manager to connect to your MariaDB RDS instance.

9. Interact with the Database: After successfully connecting, you'll be able to interact with your database using HeidiSQL's user-friendly interface. You can execute SQL queries, manage tables, and perform various database operations.

